

Food Flow Kitchen Feasibility Study



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The Food Flow Project 2014

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Photo credit: Robin Newman

Executive Summary

This feasibility analysis is part of the larger Food Flow project. Food Flow seeks to improve the flow of healthy food to the community food sector (drop in centres, multiservice agencies, shelters, supportive housing and other meal providers across the city). The Food Flow project, in collaboration with many partners, including the Community Food Procurement project at Toronto Public Health, has examined and developed plans for intervening at various points in the current community food system from primary production to consumption. The partners conducted a joint research project in 2013 that resulted in several key reports and a better understanding of the community food sector in general¹.

At the production end of the food system, the project has begun to address partnerships with the farm sector and meat processing sector to access surplus products during the local harvest (seconds, products with cosmetic defects or sizes that cannot be sold at retail). The 2013 research also identified key bottlenecks for the agencies that receive donated food, including the uneven flow of donations (large volumes of one thing, not enough of other things). The Food Flow Kitchen is a commercial processing kitchen that will process surplus from all community agency streams, and provide training and workplace opportunities to sector staff and clients. The Food Flow Kitchen will be a social enterprise that combines social purpose, community development and community engagement goals with a financially viable business.

The Feasibility Study examines three possible scenarios for a commercial processing kitchen to address the needs of the community food sector. The three scenarios tested here are:

1. A new space based near partner agencies
2. Rental in a partner's existing space
3. A mobile food processing kitchen

The feasibility study, using revenue and expense estimates from the 2013 research, new Food Flow research on pricing, and estimates from similar feasibility studies, assesses the viability of the three scenarios. In general, a net surplus of 2% was considered sufficient. Margins in food businesses tend to be very low; in this case, the organization may be a non-profit with goals of achieving only enough profit to provide some buffer for unforeseen circumstances. Revenue targets were compared to projected revenues from 10% of agencies (20-30 agencies) based on limited data from 2013. Year 1 revenue projections are very conservative, and can be increased with more information (during business plan development). It is also unlikely that food businesses break even in the first year; a business plan should provide strategies to cover the development period (probably two years). Income forecasts are provided to show the break even point for each scenario.

The feasibility study also assesses non-financial impacts (social, environmental, community, health). The Food Flow Kitchen will be a social enterprise and have a social purpose as well as a

¹ See the following link for these reports: <http://parkdalecommunityeconomies.wordpress.com/2014/02/11/just-released-community-food-flow-final-research-report/>.

healthy bottom line. The feasibility study model has been expanded, reflecting a cost-effectiveness approach, to test outcomes beyond net profit and growth targets.

Table A: Revenue/ expenses by scenario

Scenario A: New space		
Item	High	Low
Total operating expense (Year 1)		
TOTAL EXPENSES	\$480,855	\$260,302
TOTAL REVENUE required	\$490,668	\$265,614
Net surplus	2.00%	2.00%
Scenario B: Partner space		
Total operating expense (Year 1)		
TOTAL EXPENSES	\$335,000	\$192,500
TOTAL REVENUE	\$341,837	\$196,429
Net surplus	2.00%	2.00%
Scenario C: Mobile kitchen		
Total operating expense (Year 1)		
TOTAL EXPENSES	\$173,783	\$149,500
TOTAL REVENUE	\$177,330	\$152,551
Net surplus	2.00%	2.00%

Start up expenses are much lower for a shared space, assuming that the rental includes access to an existing kitchen that already has most of the necessary equipment.

Table B: Start up expenses summary

Summary start up expenses		
	High	Low
Scenario A: new space	\$124,935	\$90,450
Scenario B: partner space	\$62,886	\$46,651
Scenario C: mobile kitchen	\$114,386	\$72,151

The partner space model probably has the highest and broadest impact on the additional goals. In particular, positive impact of a community food sector training and mentoring processing enterprise would be higher; the mobile kitchen would be limited by space in the amount of training it could provide. The mobile kitchen has the potential to have more impact in community engagement and awareness rather than specific training and job placement.

In general, it was found that the mobile kitchen has fewer operating expenses and the greatest likelihood of a positive net balance after the first year. The study found that the greatest risk is probably undertaken in the mobile kitchen scenario, since it is a largely untested model for this particular use. Mobile kitchens are not uncommon, but they are used as moving restaurants, not as processing units. Risk mitigation recommendations are provide in the “Challenges and Solutions” section.

A timeline for development of the kitchen is provided (page 35). A table of additional funding options for the kitchen is also provided (page 31). Some of these activities have already been funded and are underway. Resources, equipment suppliers and a detailed price comparison for the most commonly purchased items are provided in the Appendices.

The study indicates that a combination of the mobile kitchen and partner space would have the most impact and address a range of needs; however, with limited funding, the partner space is probably the most effective approach, with the least risk and the most impact across a range of measurements. Overall, a commercial processing kitchen that meets the identified needs of the sector, and has a social impact as well, is clearly a viable enterprise and fills a clear demand. The ability to minimally process and store surplus donations, farm seconds, and other large volumes can increase access at agencies to healthy food, reduce the volatility of their supply, and provide important nutritional and health impacts for the agencies' clients and members.

Overview

The Food Flow Kitchen project responds to significant challenges in the flow of food from producers to social service agencies. Fresh food, especially locally grown food, tends to be available unevenly; acres of tomatoes are all ready at once in late summer, bins of apples are harvested in the fall. Numerous innovations in technology have sought to control this with season extension such as greenhouses, and storage extension such as climate controlled apple storage (which prevents decomposition until it is opened). At the agency end of the food flow stream, donations can be uneven as well, with large quantities of one item that the agency (or food bank distributor) cannot store or handle.

The approach to this assessment is inspired by the cost-effectiveness study model. A modified version of this approach gauges the risks, challenges, revenue/ expenses, range of benefits and impact of the project in order to move to the next stage of development. Cost-effectiveness approaches tend to be used in situations where a pure dollar figure assessment of benefits is inappropriate or does not capture all the effects of a proposed change (for instance, in the area of health care, in which quality and length of life may be outcomes of a project). This report identifies a range of potential benefits and impacts from the project, identifies measures of success that can be used to track the impact, and assesses the project's effectiveness based on a range of outcomes. Revenues and expenses are also assessed to determine the financial viability of the proposed project.

The loss of small to mid-scale processing in rural and urban centres in Canada has led to a severe bottleneck in the flow of food². New infrastructure and expertise is needed to build a resilient local food system again. The Food Flow Kitchen provides a social enterprise solution to the problem. The project will establish a training and production kitchen that can respond quickly to processing and culinary needs in the community food sector. The facility will provide the infrastructure to process bulk purchases and donations from farmers, food bank distributors, etc. The project will also create high quality jobs, provide workplace training and engage marginalized groups in workplace opportunities.

The Food Flow (FF) Kitchen can offer the following services:

- Training and workplace opportunities for marginalized groups
- Access to a network of work opportunities in hundreds of agencies as well as the food sector
- Secondary (basic) processing for surplus from farms or large volume donations to agencies or distributors
- Incubation for food-related social enterprises
- Key replicable infrastructure for strong local economies and food systems
- Improved nutritional quality of food available to the community food sector
- Increased access to locally grown fresh food for the community food sector

² See Carter-Whitney and Miller. 2010. *Nurturing Fruit and Vegetable Processing in Ontario*. Metcalf Foundation Food Solutions papers.

Situated between the donated and purchased streams of food to agencies, the project presents unique problems for feasibility testing. As a non-profit with access to grants, donors, and strategic charitable partners, the project can draw on these resources for start-up, expansion and development of some aspects of the project. However, the intention is to create a self-sustaining organization at the core. The following assessment examines three possible scenarios to the development of a sustainable processing kitchen that meets the community food sector's needs. The scenarios include:

1. Scenario A: A new independent kitchen space
2. Scenario B: A kitchen based at and accessing existing infrastructure from a key partner
3. Scenario C: A mobile processing kitchen

These scenarios will be explored and tested in turn. The report also identifies a range of goals and value-based benchmarks for success that can expand the definition of success for an enterprise with social and environmental purposes beyond net profit. The range of measurements should guide decision-making as much as the net profit goals.

The Food Flow Kitchen project developed as a key outcome of a joint research project conducted in 2013 by the Parkdale Activity Recreation Centre (PARC) with funding from the Ontario Trillium Foundation, and Toronto Food Strategy (TFS) at Toronto Public Health with funding from the Province of Ontario Healthy Communities Fund and the Public Health Agency of Canada³. For the PARC project, the development of the Food Flow Kitchen represents a pilot project funded by the Ontario Trillium Foundation. The 2013 research project assessed over 350 agencies in the City of Toronto that provide meals to people facing food insecurity. Overall, the research team estimated that the sector provides over 6.5 million meals and spends around \$29 million annually, mostly at retail stores, to supplement donations through Daily Bread, Second Harvest, North York Harvest and various local donors. In addition, the Student Nutrition Program spends another \$17 million to provide healthy meals to children in need.

The research found that although a considerable amount is spent each year on food, the expenditures are not well-organized. Many purchases are last-minute or unplanned. Food programs tend to rely on non-food staff, taxis or personal vehicles, unpaid staff-time or untrained volunteers. The spending power has not been organized to command volume purchases or to meet the specific needs of a sector providing millions of meals annually. The research project has catalyzed several important responses, including the development of aggregated purchasing through an online website to access fresh and local food, storage solutions like shared freezer space for meat, and strengthened alliances with the farm sector to access surplus more efficiently. The Food Flow Kitchen addresses another node in the local food system—processing. The project can reduce waste, create jobs, and improve the nutritional quality of food at agencies.

The Food Flow project has developed a network of partners and allies since its inception in November 2012. Funding has been gratefully received from the Ontario Trillium Foundation and the Toronto Enterprise Fund. A similar project shared data and results and was sponsored and funded through Toronto Food Strategy (Toronto Public Health). In-kind support from PARC for

³ See <https://parkdalecommunityeconomies.wordpress.com/2014/02/11/just-released-community-food-flow-final-research-report/> for reports from the 2013 research.

infrastructure, staff support and management oversight has been crucial to the project's success. In addition, the project has received generous in-kind support, insights and planning from the Community Food Flow Advisory Group as well as a broad network of West Central Toronto agencies and beyond. Around one hundred agencies generously provided their time and expertise during in-depth interviews across the city. The Food Flow project currently provides a monthly newsletter to hundreds of agencies and supporters to maintain the network and ensure a broad impact for the projects pilots and development process.

PARC is a multi-service agency operating various community programs and services for survivors of mental health ailments, the homeless, the disabled and those with few resources. PARC actively works to re-skill their members. PARC has a record of the successful development of social enterprise with Silver Brush, as well as the collaborative development of the West End Food Co-op community cannery, and the Co-op Cred program which provides food-related workplace opportunities at local agencies for marginalized people.

Goals and Values of Food Flow Kitchen Project

The general categories of goals include business viability (measured in terms of sales, profits, and various financial benchmarks). The Food Flow Kitchen also addresses a number of goals and values, some of which are difficult to quantify in dollar terms. Categories of impact include goals of environmental impact reduction, community development, employment development and job creation, education, and physical and mental health.

The benefits of the Food Flow Kitchen project, which can represent areas for targets and measurements of success are summarized below. Unlike triple bottom line assessments, which tend to measure the impact of business activities in one of three categories (social, economic, environmental), this approach accounts for the multiplying effect of activities. That is, some activities may have positive benefits of more than one kind. A program that provides training, mentoring and job support for marginalized groups provides employment development and creation, but also affects the education goal. A successful enterprise that is responsive to the needs of community organizations can be both a viable business and a tool for community development.

The Food Flow Kitchen seeks to:

- Create a viable and self-sufficient enterprise (business viability)
- Reduce waste at agencies receiving large donations and for farmers with seconds (environment)
- Create a strong local business that is community-based and responsive to the needs of local organizations (business viability, community development)
- Provide new opportunities for people who are street-involved, marginally housed, and low income or face other employment barriers (employment development/ creation, education)
- Create a program for training, certification (food handling) and placement for community chefs working at social service agencies (education)
- Provide the infrastructure for the incubation of other community-based and social enterprises (community development)
- Provide fresher, healthier options for the community food sector (health)
- Reduce environmental impact of long-distance food and food waste (environment)

Food Flow Kitchen scenarios

Three basic scenarios can meet the goals of the project. Each has pros and cons and depends on a unique governance and financial structure. The following section provides details on each option.

1. New social enterprise in dedicated space

The Food Flow Kitchen can be located in its own space, probably in the Parkdale area to be close to established partners (PARC, West End Food Co-op, the Co-op Cred program, Greenest City, etc.). The scenario establishes a dedicated space with equipment appropriate to the scale and plans of the project. Participants and staff would have access at all times, and would be able to adjust use to need without regard for other program schedules. The scenario probably means increased start up costs. As an independent operation, the project would need to provide infrastructure from the beginning (phone system, copy machine, etc.). In a partner scenario, these items can be shared through a percentage based on use.

Advantages:

The scenario provides some independence from the vicissitudes of expansion and contraction at a host organization (the risk of eviction either because the host needs the space or because the host is moving or collapsing). Access to a Parkdale location means proximity to similar organizations that create community support and awareness, access to clients from partner programs within walking distance, as well as access to potential renters as other food-based initiatives develop.

Disadvantages:

Disadvantages include the risk of renting from a conventional landlord (rather than a charitable or non-profit sympathetic host). A new space may mean increased operating costs compared to expenses with partners if renting within an existing organization.

In any scenario, rents may increase, but a supportive host is more likely to be willing to work with the new enterprise to ease the impact. In addition, Parkdale rents are high and increasing. The Food Flow Kitchen is a small-scale industrial project with no retail operation, and therefore does not require easy access to customers. A Parkdale location is less likely to have good access to parking and delivery areas for commercial vehicles. The latter considerations suggest that a dedicated space would be better established outside Parkdale but nearby (Etobicoke, for instance).

2. Rental from existing and supporting partner

The Food Flow Kitchen in this scenario would rent space and access to existing equipment from a supporting partner such as Daily Bread.

Advantages:

Daily Bread and other partners have extensive well-maintained equipment and kitchen space that is not in use at all times; the rental could be by a half day or even hourly to give the enterprise flexibility and responsiveness to agency need and availability of surplus. Daily Bread already has an extensive training program, and various options exist to connect the new social enterprise with the existing program. Daily Bread also has extensive expertise in running a commercial kitchen, as well as existing unused storage space (walk-in freezers and coolers).

The rental agreement is likely to be a better arrangement than the enterprise could get with a commercial landlord; even with standard rents, other aspects in the lease may show a better awareness of the issues and needs of a new social enterprise, and more willingness to accommodate specific requirements. Daily Bread's values are similar to the Food Flow Kitchen, so there would be less chance of clashes of values and misunderstandings than the project would risk in a commercial rental space.

Disadvantages:

The disadvantages of renting in a partner space include the need to negotiate and navigate use times and agreements around maintenance, clean-up, etc. In addition (see above), the host may expand and need the unused space for their own program, or, particularly if they are a small operation, they may be forced to close and leave the enterprise without space.

3. Mobile processing kitchen

The mobile processing kitchen solution is less familiar in the North American landscape. Mennonite groups have used mobile canneries for decades mostly for meat. The North Dakota Department of Agriculture has launched a very large one for local farmers. Its success is still uncertain (Goode, personal communication). These are rural projects; an urban kitchen faces additional challenges around permitting, space to set up, etc. Goode recommends the project consider a much smaller vehicle that has the basic canning items for high acid canning, and perhaps a dehydrator and freezer. The start-up capital would be much lower (their unit alone cost 60K, and requires specific driving expertise and insurance).

Advantages:

A mobile project addresses some of the key issues that a fixed kitchen cannot solve easily. If agencies receive large donations they cannot handle, processing in a fixed kitchen would require moving the product to Food Flow's Kitchen to access the equipment. Another approach would be a processing "flying squad" of community chefs able to go to agency kitchens when needed to respond to processing or even cooking needs. The mobile kitchen project could be folded into either of the other two to add revenue and flexibility to the programs. The mobile project might provide capacity to process on-farm, and increase access to local fresh farm seconds.

Disadvantages:

As an innovative solution, the project would have few similar projects to draw on for recommendations and cautions. Permitting of food-related projects in Toronto has proven extremely challenging in the case of the food truck projects. Depending on the size, parking for many agencies might constitute a problem. The size of the mobile unit would also limit output as well as the number of trainees that could be accommodated.

Matching Goals and values

Each of the aspects of the Food Flow Kitchen meets one or several goals of the project. The Table 1 below shows the overall impact of the project.

Table 1: Impact on goals and values of project

Goals and values of project	Categories of impact	Business viability (long-term)	Business viability (short-term)	Employment development/creation	Positive impact on planet	Community development	Education	Health (physical and mental)
Create a stable and self-sufficient enterprise		✓	✓					
Reduce waste in food products at agencies who receive large donations and farmers with seconds					✓			
Create a strong local business that is community-based and responsive to the needs of local organizations			✓			✓		
Provide new opportunities for people who are street-involved, marginally housed, and low income or face other employment barriers				✓		✓		
Create a program for training, certification (food handling) and placement for community chefs working at social service agencies				✓		✓	✓	✓
Provide the infrastructure for the incubation of other community-based and social enterprises						✓		
Provide fresher, healthier options with local food								✓
Reduce environmental impact of long-distance food and food waste					✓			

Tracking Success

The goals and values of the project can be matched with indicators and measurement units that allow the cost-effectiveness of the operation to be calculated and tracked.

For instance, the goals shown in Table 2 can be tested and tracked with the associated indicators. Benchmark amounts for final indicators should be identified subsequently in the project business plan.

Table 2: Indicators of progress

Goal of project (expanded)	Indicator
Create a sustainable social enterprise that addresses growing market demand for local products	▪ Sales amount per week/ month/ year ▪ Gross and net surplus ▪ Market research and long-term plans ▪ Revenue and expense targets
Create a strong local business that is community-based and responsive to the needs of local organizations	▪ # of local stakeholders on board ▪ # of people consulted in town meetings, etc.
Provide new opportunities for people who are street-involved, marginally housed, and low income	▪ # of people accessing project ▪ demographics of participants ▪ year to year impact of participation (long-term)
Provide new workplace opportunities for people facing employment barriers	# of employment opportunities created
Provide mentoring and training for marginalized groups to access an expanding sector of Ontario's economy (food processing and preparation)	# of participants able to access related employment after program # of transferable skills to apply in this or related sectors
Create a program for training, certification (food handling) and placement for community chefs working at social service agencies	▪ Size of program (dollars, number of participants) ▪ Number of placements
Provide the infrastructure for the incubation of other community-based and social enterprises	▪ # of templates ▪ # of outreach events and communications about project ▪ # of other groups assisted
Maintain a balance between financial and social goals	▪ Promotion and publication of goals ▪ Integration of goals into all long-term plans ▪ Integration of goals into evaluation process (strategic and staff)
Reduce waste in food products sector (at agencies receiving large donations and for farm seconds)	▪ Amount of waste saved by weight ▪ Amount of waste rerouted (to compost)
Increase access to healthy food	▪ Amount of sales and distribution that is fresh/ local food ▪ # of agencies (customers) ▪ Nutrition content of food offered
Increase access to fresh local food	▪ Amount of sales and distribution that is fresh/ local food ▪ # of local suppliers ▪ # of agencies (customers) ▪ Amount purchased per supplier
Develop high quality job opportunities	▪ Level of skills learned (basic to highly technical) ▪ Pay scale ▪ Range of benefits available ▪ Number of hours average per worker
Provide positive impact on local economy	▪ # of local secondary suppliers (equipment, ingredients, etc.) ▪ # of viable incubated businesses

Financial scenarios

Revenue

The 2013 research indicated a strong demand for fresh product; however, actual purchasing practices show a reliance on processed products. The initial research did not explore fully the amounts of different products that were purchased. Eight agencies provided invoices covering one to two months in the fall of 2013 for comparison. This showed a wide variation in which the only common denominators seemed to be dairy, eggs, canned tomatoes and apples. However, the product list was not a focus of the research. At this point, the limited knowledge means that revenue projections must be very conservative. Of 350 agencies purchasing food, the projections are based on around 10% of them buying from the new distribution option each week.

Next steps will include narrowing the plan down to one scenario or a combination of them, and building a business plan with more detailed information around market opportunities and costs (start up and operating). The feasibility scenarios are designed for decision-making purposes, and represent a basic overview only.

The expectation of limited sales at first increases the risk factor for the new space Scenario A, where the organization operates independently of other distributors. Low sales would create more of a challenge; the organization would still need to support some or all of the distribution infrastructure (truck, driver, invoicing).

Scenario B (Partner space) would reduce this risk if the FF Kitchen could share or purchase space on the partners' trucks when needed. If successful, this scenario creates a later challenge if the sales expands and the partner can no longer handle deliveries of the volume. However, later risk is probably safer because the FF Kitchen could predict the challenges and would have time to take steps to a solution.

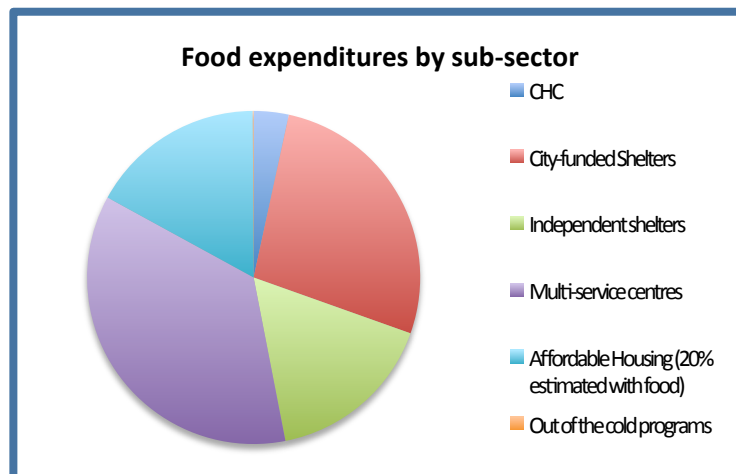
The mobile kitchen Scenario C can create a sufficiently flexible model to avoid these problems; the manager could also be the driver, since the kitchen would go where it was needed. The model essentially removes the need for distribution in some cases, although in processing amounts larger than the host agency can use, the FF Kitchen would still require distribution.

Year 1 revenue projections were cross-linked to revenue targets to indicate the difference for each scenario between targets (to cover expenses) and estimated project sales (see below on p. X). The business plan should develop marketing plans to ensure the targets are met.

Market Opportunities

The 2013 Community Food Procurement research identified about \$29 million in food expenditures annually by the sector. The expenditures are not divided equally: multi-service centres provide more meals on much lower budgets than the other categories of agencies. Table 3.1 shows a market overview by sub-sector.

Table 3.1: Market overview



Community Food Sector Needs and Solutions

The City of Toronto's 2008 Report *The State of Toronto's Food* provides an overview of the challenges for the community food sector:

- "The average food item sold in Toronto has traveled nearly 4500 km"
- "The food system is responsible for 30% of pollution and greenhouse gas emissions"
- "There are only three days worth of fresh food in the city at any time"
- "Food prices in Toronto are lower than in most other countries but low wages, low social assistance rates and the high cost of housing still cause many to go hungry"

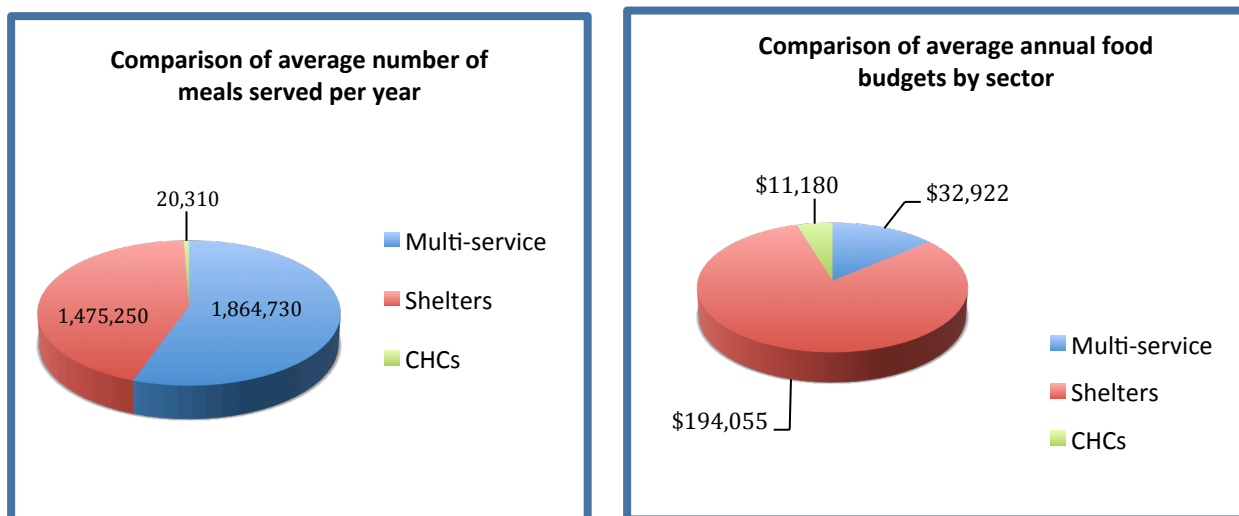
In addition the report cites high levels of nutrition related diseases that cost public money and are preventable through improvements to the local and regional food system, including obesity, diabetes and heart disease. Over one in three Toronto children are now reported to be overweight or obese (City of Toronto 2008)⁴.

The two research reports (*Finding Food* for Toronto Food Strategy and *Reassembling Community Food Flow* for PARC) summarize the characteristics of the sector and recommend solutions for the various challenges. In general, the sector includes a range of agency types, although multi-service agencies (with drop-in meal programs and many other functions) tend to be the most numerous. They provide the most meals and receive the least funding from any source (public or private): see tables above. A rough estimate places annual expenditures around \$29 million. The estimate includes drop-in centres, shelters, community health centres, church meal programs and a percentage of supportive and affordable housing expenditures. It does not include institutions (hospitals, etc.), or the School Nutrition Program, which alone spends \$17

⁴ The 2011 and 2012 feasibility studies for the West End Food Co-op Kitchen have provided some important background and input for this project.

million annually on food. Table 3.2 provides more detail (number of meals, average budgets) for a sector overview.

Table 3.2: Sector overview



Agencies face numerous barriers to providing fresh, healthy food. Few of them have dedicated or sufficient food program staff. Many programs are run largely by volunteers, or by managers from other programs. Most organizations do not have healthy eating guidelines to help non-food program staff purchase to meet nutritional needs of clients. Few funding organizations, either public or private, provide funds for direct service such as meal provision. Agencies recognize the impact of hunger on a client's ability to benefit from other services, access employment, think clearly, or manage housing. Funding for these expanding programs tends to be siphoned from other programs.

With limited funding, staffing, and organizational priority, food tends therefore not to be a focus for agencies. As Tarasuk's research showed (2003), meal programs tend to offer insufficient nutrients to meet a client's daily needs. This can be due partly to lack of knowledge. It is also attributable to the dependence on donations. Multi-service agencies can have budgets as low as .15/ plate (or even \$0); this means they cannot control the food they have access to for meal provision. Daily Bread and other food bank distributors work hard to provide fresh healthy food whenever possible.

Primary processed products with minimal additives such as what the FF Kitchen will produce can increase the nutritional value, and expose clients to fewer chemical additives, salt and sugar. In addition, the FF Kitchen can intervene in some categories that tend to be filled with unhealthy choices due to lack of access and convenience. This includes breakfast foods that are high in sugar, corn syrup and salt, and between meal snacks that are sugary and low in nutritional value. Some studies show that local food retains its nutritional value better than food that has traveled thousands of miles (i.e., most of our imported food). Certainly, local fruit and vegetable growers can harvest products at a riper stage since the danger of bruising is less and the distance to the customer is shorter. Ripe harvesting improves the nutrient value (the plant has time to fully develop vitamin and mineral content) and also taste, as the sugars develop only when the product

is ripe. Local food tends to come from unsubsidized small to mid-scale farms, and can be more expensive than imports. Purchasing practices will need to balance these considerations.

Local food is not an explicit goal for most agencies; cost and convenience are the driving factors in purchase decisions. However, Food Flow projects like the FF Kitchen recognize the potential to solve problems and meet the needs of the sector through local food procurement that supplements and replaces the current reliance on imports.

Additional market research is needed, but market opportunities have been identified during research for this project and similar research for the West End Food Co-op during the development phase. Diversified revenue may be key to success in the first few years, matching the diverse needs of the 350 potential agency customers.

Several significant opportunities have been identified and calculated in the Revenue Projections Table 4 below. Most of the agencies use processed tomatoes at least once per week; this is a straightforward high acid canning process that fits the abundance of the harvest cycle.

Breakfast and snacks are a particular challenge for agencies; often it is provided by non-food staff, and consists of coffee and convenience foods like supermarket muffins. Baking is rarely undertaken at agencies. A small bakery operation at the FF Kitchen can meet the need for healthier breakfast options and between meal snack options as well. Homemade muffins and scones can be healthier simply because they tend to have less sugar and salt, and no corn syrup. Granola bars are simple to make, transportable, and need no refrigeration. Likewise, dehydrated fruit can provide a nutritious snack that clients can eat later, and that can be stored at room temperature.

Agencies seem to rely on frozen vegetables; a commercial kitchen can meet some of these needs, and provide for a quick response to surplus when it is not possible to process the products, or the product is low acid (which requires pressure canning). Freezing can be used for nutritious soups and sauces as well.

The revenue projections assume some income for processing large amounts of donated items; this fee is set at \$25/ hour to the agency or organization providing the donated goods, and does not fully recover the costs of processing⁵. The social enterprise is designed to be able to provide this much needed response for large donations, while maintaining financial viability through other services.

The FF Kitchen team can provide processing workshops at the FF Kitchen or at other agencies; these would focus on training chefs in basic storage techniques—hot water bath canning, perhaps oven dehydration, freezing. At this point, there is no additional revenue projected from training, but at some point this aspect of the project may generate fees or other income to support the educational aspect. A small amount of revenue is projected to come from grant income, particularly in the first few years of development.

⁵ The hourly rate is drawn from similar projects such as FoodShare's Kitchen Incubator project and the New City Market draft business plan (http://www.newcitymarket.org/business_plan/commercial-kitchen-revenues/).

Table 4: Revenue Projections by Type

Item	Quantity monthly	Quantity annually	Price	Unit	Total revenue
Processed tomatoes	960	11520	\$1.35	796ml	\$15,552
Muffins	200	2400	\$3.67	540g	\$8,808
Bread	144	1728	\$3.00	loaf	\$5,184
Frozen mixed vegetables	160	1920	\$1.11	398 ml	\$2,131
Dehydrated apples	50	600	\$7.67	lb	\$4,602
Dehydrated other fruit	25	300	\$7.00	lb	\$2,100
Processed high acid other	1500	18000	\$3.00	500 ml	\$54,000
Processing donated items	64	768	\$25.00	hour	\$19,200
Workshops	2	24	\$400.00	session	\$9,600
Grant income		0			\$30,000
Total product sales					\$92,377
Total other income					\$58,800
TOTAL					\$151,177

Expenses

Start up expenses and operating expenses were identified for all three scenarios. Scenario C (mobile kitchen), although similar in capital and start up requirements, was the closest to matching revenue projections to expenses. The lack of fixed infrastructure reduces the expenses. In each case, a high cost/ high revenue and a lower cost/ revenue scenario was analyzed. The estimates of costs and revenues are drawn from two feasibility studies completed for the West End Food Co-op Kitchen and Cannery, two other feasibility studies for similar projects, and quotes from various equipment suppliers. In addition, PARC and Daily Bread provided expense figures for their kitchen programs. In 2014, additional research from Food Flow identified the standard prices on the most frequently purchased items (see table in Appendix C); these were used in the revenue projections. Revenue reported below is the revenue target necessary to cover costs in each scenario; in Scenario A and B, Year 1 revenue projections (based on existing research) do not cover costs.

Note that revenues in Table 5 are a target goal designed to achieve approximately 2% net surplus. Actual revenue projections based on known (conservative) market potential is provided in Table 4. The difference is summarized in Table 10: Revenue Projections vs. Break-even Requirements.

Table 5: Summary of Revenue and Expenses (by scenario)

Scenario A: New space		
Item	High	Low
Total operating expense (Year 1)		
TOTAL EXPENSES	\$480,855	\$260,302
TOTAL REVENUE required	\$490,668	\$265,614
Net surplus	2.00%	2.00%
Scenario B: Partner space		
Total operating expense (Year 1)		
TOTAL EXPENSES	\$335,000	\$192,500
TOTAL REVENUE	\$341,837	\$196,429

Net surplus	2.00%	2.00%
Scenario C: Mobile kitchen		
Total operating expense (Year 1)		
TOTAL EXPENSES	\$173,783	\$149,500
TOTAL REVENUE	\$177,330	\$152,551
Net surplus	2.00%	2.00%

The equipment projected for the initial start up provides for a basic operation that can do basic steam kettle processing (hot water bath), freezing, dehydrating and baking. These items apply mostly to the New Space Scenario A, but the mobile kitchen, due to its novelty, may incur additional costs to replace standard equipment (like deep fryers) with specialty processing equipment like a dehydrator. Note that ownership of the equipment (Scenarios A and C) should be considered a risk, as the project would have to maintain and calculate depreciation on these. After a few years their resale value would probably be minimal. Sources for kitchen equipment are listed in Appendix B. The following tables show the estimated equipment expenses. These expenses are largely applicable in the “new scenario A (new space). Table 6, 7.1 and 7.2 show the start up expenses for each scenario. It is assumed in Scenario A that, unless specific funding for renovations is acquired, the new space will be an existing kitchen that needs only the processing equipment and minor repairs or changes to be functional.

Table 6: Start up equipment expense (new and partner space)

Basic equipment item	Unit cost (CFE)	Unit cost 2 (Nella/ WEFC 2011)	Number	Average	Total
2 Burner induction cooker	\$3,000	\$1,500	3	\$2,250	\$6,750
Walk in freezer 12' x 10' x 8' high	\$8,300		1	\$8,300	\$8,300
30 QT dough mixer	\$3,800	\$1,000	1	\$2,400	\$2,400
24" pot 2 compartment sink	\$800	\$499	1	\$800	\$800
Grease trap	\$420		1	\$420	\$420
2 door cooler	\$2,190	\$3,000	1	\$2,595	\$2,595
Electric convection oven	\$3,200	\$2,899	1	\$3,050	\$3,050
Hand wash sink	\$120	\$149	1	\$135	\$135
Food dehydrator	\$700		1	\$700	\$700
Tilting kettle	\$7,000		2	\$7,000	\$14,000
4 quart food processor	\$1,700		2	\$1,700	\$3,400
Large juicer	\$1,250		1	\$1,250	\$1,250
Total Basic items					\$43,799
Other equipment or options					
18" pot 3 compartment sink	\$750		1	\$750	\$750
Electric convection oven	\$4,500		1	\$4,500	\$4,500
Tilting skillet electric	\$13,000		1	\$13,000	\$13,000
Total other equipment					\$18,250

Table 7.1: Non-equipment start up expenses for new and partner space

Start up salary expenses	High	Low
Development manager	\$43,000	\$40,000
Kitchen manager	\$50,488	\$33,426
Sub-total	\$93,488	\$73,426
MERCs	\$10,284	\$8,077
Totals salary	\$103,772	\$81,503
6 month total salary	\$51,886	\$40,751
Other expenses		
Legal/ permitting	\$500	\$500
Marketing	\$500	\$500
Insurance	\$500	\$400
Miscellaneous kitchen	\$2,000	\$1,500
Rent	\$7,500	\$3,000
Total other expenses	\$11,000	\$5,900
Total start up new space (with equipment Table 6)	\$124,935	\$90,450
Total start up partner space (with equipment Table 6)	\$62,886	\$46,651

Table 7.2: Non-equipment start up expenses for mobile kitchen

Mobile Kitchen start-up costs	High	Low
Mobile kitchen unit	\$60,000	\$30,000
Legal/ permitting	\$1,500	\$500
Marketing	\$500	\$500
Insurance	\$500	\$400
Total start up expenses	\$62,500	\$31,400
Start up salary expenses		
Development manager	\$43,000	\$40,000
Kitchen manager	\$50,488	\$33,426
Sub-total	\$93,488	\$73,426
MERCs	\$10,284	\$8,077
Totals salary	\$103,772	\$81,503
6 month	\$51,886	\$40,751
Total start up expenses mobile kitchen	\$114,386	\$72,151

Operating expenses are designed to start as low as possible while still providing for a trainee crew, one of the key goals of the project. For the low cost scenarios, in general the kitchen manager is the main staff-person, with some administrative backup. Operating expenses are

highest for the Scenario A (new space), lowest for the Scenario C (mobile kitchen). Operating expenses in Scenario C are reduced by not paying rent and other infrastructure costs of maintaining a fixed space. However, some rental of freezer/ cooler storage and office space might still be necessary, increasing these expenses. Scenario A would be even higher in start up and operating costs if the project had its own or leased trucks and hired a driver (estimated at \$82,800 annually based on comparable Toronto distributor costs).

Table 8: Estimates of cost of trucking⁶

Truck expenses	High	Low	
Freight charge (per pallet)	\$30,000	\$15,000	30-50/ pallet
OR			
Truck rental	\$24,000		
Fuel	\$9,600		
Km charges and reefer	\$4,200		
Driver	\$45,000		
Annual cost of leased truck	\$82,800		

Initially at least, it makes more sense while sales develop to hire a common carrier or partner to deliver product for a per pallet price. The latter is estimated at \$30-50 depending on volume, or 15-30K annually; this is estimated lower for partner distribution as it is expected that routes would be shared.

Revenue Requirements and Operating Expenses

The following tables (9.1, 9.2, 9.3) provide estimates of expenses for a high cost and low cost budget for each scenario. The tables show the corresponding revenue requirements to generate a 2% net surplus. These revenue requirements are compared to revenue projections in Table 10 (based on current information and conservative estimates).

Table 9.1: Revenue requirements and operating expenses for Scenario A: New space

Expenses		
	High (4000 sq. ft)	Low (2000 sq. ft)
COGS	\$178,000	\$92,000
Operating		
Utilities	\$21,749	\$10,875
Other occupancy	\$37,100	\$18,550
Office/ admin/ marketing	\$21,506	\$15,878
Depreciation (included in occupancy)		
Total operating	\$80,355	\$45,302
Distribution		
Agency coordinator	\$35,000	\$10,000
Truck rental		
Transportation costs (fuel)		
Insurance		
Driver		
Freight (pallet price)	\$41,600	\$20,000
Total distribution costs	\$35,000	\$30,000
Rent annual	\$90,000	\$36,000

⁶ These numbers were generously provided by an existing food distributor in Toronto.

Salaries		
Kitchen manager	\$50,500	\$45,000
Assistant manager	\$35,000	
Other staff		
Trainee wages	\$12,000	\$12,000
Total salaries	\$97,500	\$57,000
TOTAL EXPENSES	\$480,855	\$260,302
TOTAL REVENUE required	\$490,668	\$265,614
Net surplus	2.00%	2.00%

Table 9.2: Revenue requirements and operating expenses for Scenario B: Partner space

Expenses		
	High (8000 sq. ft)	Low (2000 sq. ft)
COGS	\$120,000	\$64,000
Operating		
Utilities	\$12,000	\$10,000
Other occupancy		
Office/ admin/ marketing	\$500	\$500
Depreciation (included in occupancy)		
Total operating	\$12,500	\$10,500
Distribution		
Agency coordinator	\$15,000	\$10,000
Truck rental		
Transportation costs (fuel)		
Insurance		
Driver		
Freight (pallet price)	\$30,000	\$15,000
Total distribution costs	\$15,000	\$25,000
Rent	\$90,000	\$36,000
Salaries		
Kitchen manager	\$50,500	\$45,000
Assistant manager	\$35,000	
Other staff		
Trainee wages	\$12,000	\$12,000
Total salaries	\$97,500	\$57,000
TOTAL EXPENSES	\$335,000	\$192,500
TOTAL REVENUE	\$341,837	\$196,429
Net surplus	2.00%	2.00%

Table 9.3: Revenue requirements and operating expenses for Scenario C: Mobile kitchen

Expenses		
	High	Low
COGS	\$48,000	\$44,000
Operating		
Utilities		
Mobile unit operating costs (ND model)	\$14,183	\$12,000
Other occupancy: parking	\$1,200	\$1,200
Office/ admin/ marketing	\$500	\$500
Depreciation (included in occupancy)		
Total operating	\$15,883	\$13,700
Distribution		
Agency coordinator	\$15,000	\$10,000
Transportation costs (fuel)	\$1,200	\$800
Insurance	\$1,200	\$1,000
Driver		
Freight (pallet price)		
Total distribution costs	\$17,400	\$11,800
Rent	\$0	\$0
Salaries (with MERCS)		
Kitchen manager	\$50,500	\$45,000
Assistant manager (P/T)	\$30,000	\$23,000
Other staff		
Trainee wages	\$12,000	\$12,000
Total salaries	\$92,500	\$80,000
TOTAL EXPENSES	\$173,783	\$149,500
TOTAL REVENUE	\$177,330	\$152,551
Net surplus	2.00%	2.00%

Year 1 Revenue projections were matched to revenue targets. The difference is shown in the table 10 below. Revenue projections are conservative; the business plan should address the shortfall and provide marketing plans to achieve balanced profit and loss statements.

Table 10: Year 1 Revenue projections vs. revenue targets

Revenue target vs. Year 1 projections				
	New space			-\$114,437.24
	Partner space			-\$45,251.37
	Mobile kitchen			-\$1,374

At this point, the projections are based on an estimate of only about 10% of agencies accessing the processed products. For a sustainable enterprise, this number will be higher; 10% represents the initial expected sales only. It would be expected that after a year the project would meet necessary revenue targets. The following section explores income forecasts over 5 years to estimate the break even point for each scenario (that is, the point at which revenues are sufficient to cover expenses).

Income forecasts over 5 years

In general, food sales ventures tend to take 2-3 years to become profitable (independent of loans, grants, donors, etc.). The following Tables show the break even point over five years for each scenario. Scenario C shows the quickest turnaround (and offers the highest risk as a novel solution for most of North America). In the following tables, numbers marked in green create positive net surplus while the gray is a net loss. Both high costs and low cost estimates are shown. The lower the initial operating costs, the faster the break even point can be achieved.

Note that gray cells indicate expenses resulting in a net loss. Green shading indicates a net gain.

Table 11.1: Income forecasts over 5 years for Scenario A

Scenario A: New space					
	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue					
Sales	\$92,377	\$184,754	\$230,943	\$288,679	\$360,848
Other income	\$58,800	\$117,600	\$147,000	\$183,750	\$229,688
Total revenue	\$151,177	\$302,354	\$377,943	\$472,429	\$590,536
Expenses					
Cost of Goods Sold	\$36,951	\$120,942	\$151,177	\$188,972	\$236,214
Gross margin	\$114,226	\$181,413	\$226,766	\$283,457	\$354,322
<i>Low cost scenario</i>					
Labour costs	\$57,000	\$57,000	\$62,700	\$68,970	\$75,867
Operating expenses	\$111,302	\$111,302	\$122,432	\$134,676	\$148,143
<i>High cost scenario</i>					
Labour costs	\$97,500	\$97,500	\$107,250	\$117,975	\$129,773
Operating expenses	\$205,355	\$205,355	\$225,890	\$248,479	\$273,327
Total expenses low	\$205,253	\$289,244	\$336,310	\$392,617	\$460,225
Total expenses high	\$339,805	\$423,796	\$484,317	\$555,425	\$639,314
Net income/ loss (low)	-\$54,076	\$13,110	\$41,633	\$79,812	\$130,311
Net income/ loss (high)	-\$188,628	-\$121,442	-\$106,374	-\$82,997	-\$48,778

Table 11.2: Income forecasts over 5 years for Scenario B

Scenario B: Partner space					
	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue					
Sales	\$92,377	\$184,754	\$230,943	\$288,679	\$360,848
Other income	\$58,800	\$117,600	\$147,000	\$183,750	\$229,688
Total revenue	\$151,177	\$302,354	\$377,943	\$472,429	\$590,536

Expenses					
Cost of Goods Sold	\$36,951	\$120,942	\$151,177	\$188,972	\$236,214
Gross margin	\$114,226	\$181,413	\$226,766	\$283,457	\$354,322
<i>Low cost scenario</i>					
Labour costs	\$57,000	\$57,000	\$62,700	\$68,970	\$75,867
Operating expenses	\$71,500	\$71,500	\$78,650	\$86,515	\$95,167
<i>High cost scenario</i>					
Labour costs	\$97,500	\$97,500	\$107,250	\$117,975	\$129,773
Operating expenses	\$117,500	\$117,500	\$129,250	\$142,175	\$156,393
Total expenses low	\$165,451	\$249,442	\$292,527	\$344,457	\$407,248
Total expenses high	\$251,951	\$335,942	\$387,677	\$449,122	\$522,379
Net income/ loss (low)	-\$14,274	\$52,913	\$85,416	\$127,972	\$183,288
Net income/ loss (high)	-\$100,774	-\$33,587	-\$9,734	\$23,307	\$68,157

Table 11.3: Income forecasts over 5 years for Scenario C

Scenario C: Mobile kitchen					
	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue					
Sales	\$87,985	\$175,970	\$219,963	\$274,954	\$343,692
Other income	\$58,800	\$117,600	\$147,000	\$183,750	\$229,688
Total revenue	\$137,185	\$274,370	\$342,963	\$428,703	\$535,879
Expenses					
Cost of Goods Sold	\$35,194	\$109,748	\$137,185	\$171,481	\$214,352
Gross margin	\$101,991	\$164,622	\$205,778	\$257,222	\$321,527
<i>Low cost scenario</i>					
Labour costs	\$80,000	\$80,000	\$88,000	\$96,800	\$106,480
Operating expenses	\$25,500	\$25,500	\$28,050	\$30,855	\$33,941
<i>High cost scenario</i>					
Labour costs	\$92,500	\$92,500	\$101,750	\$111,925	\$123,118
Operating expenses	\$33,283	\$33,283	\$36,611	\$40,272	\$44,300
Total expenses low	\$140,694	\$215,248	\$253,235	\$299,136	\$354,772
Total expenses high	\$160,977	\$235,531	\$275,546	\$323,679	\$381,769
Net income/ loss (low)	-\$3,509	\$59,122	\$89,728	\$129,567	\$181,107
Net income/ loss (high)	-\$23,792	\$38,839	\$67,416	\$105,024	\$154,110

Impact assessment

The three scenarios for assessment must be tested against the goals and values established for the social enterprise; this process establishes strategic directions that are designed to look beyond financial viability. Organizations like social enterprises and co-ops can have difficulty maintaining their social purpose if the measures of success that are tracked by staff and board remain oriented to profit, with other achievements counted as secondary. The following examines each scenario in light of the overall goals of:

- Employment development/ creation
- Positive impact on planet
- Community development
- Impact on health levels
- Impact on mental health levels
- Short-term business viability
- Long-term business viability

Employment development/ creation

A primary goal for the Food Flow Kitchen is the support of access to employment for marginalized groups (both training and mentoring in job seeking, employment rights, etc.). In addition, the project seeks to provide workplace opportunities. The focus on job creation is well suited to small to mid-scale processing, which tends to be labour intensive. The projected employment and training impact is outlined below.

Table 12: Employment estimates

Position	Employment status/ hours	Notes
Kitchen Manager	F/T	slightly lower payscale in smaller operations
Assistant Manager	F/T in all high sale scenarios; P/T or none in low sale scenarios	
Agency coordinator	P/T except for large independent operation (1/3 to ½ FTE)	
Office administration	P/T except for large independent operation (1/3 to ½ FTE)	
Trainees	2-3 initially, including interns	About 65 hours per month; wages set at \$15/ hour

Accounting and book-keeping can be contracted to a third party initially for a few hours each month, increasing with the size of the operation, until in-house capacity is needed. This is part of the administrative cost in the budget. Progress in the employment and training aspect of the project can be measured by the number of participants who access training and other support services. Additional measurements of this impact include future employment and work opportunities, quality of work accessed (including arrangements for time off, flexibility of schedule, etc.), level of job skills after training, and length of time in positions (turnover).

Positive impact on planet

Purchases of seconds from farmers (that cannot be sold at retail due to cosmetic defects) reduce waste in the food system. These can be calculated by measuring of the weight of offset waste.

Purchases from local farmers reduce emissions through food miles reduction. Formulas to calculate for emissions in food distribution exist in the literature; ideally several would be applied and compared to ensure accuracy. In addition to standard recycling practices, the project can also partner with an urban agriculture organization to compost waste.

Of the three scenarios, the mobile kitchen may have the most impact simply by the potential visibility of good practices. However, this may be offset by the extra fuel use for a large trailer unit. The option to work more directly with farmers through on-farm processing also increases the positive environmental impact of the mobile kitchen, reducing waste and food transport miles.

Community development

The mobile kitchen, due to high visibility and presence at the agency may mean a greater chance to engage agency clients and volunteers, to create awareness in communities, etc. For a general educational impact, the mobile kitchen probably rates higher. However, for useful and applicable job skills that can build stronger communities, a fixed kitchen can probably host more trainees and expand to a larger program. A combination of the two could achieve the outreach results as well as concrete training and employment goals.

Measurements of this impact include the availability of templates for replication, outreach, communication and networking levels, and the incubation or support of new enterprises trained or hosted by the Food Flow Kitchen project.

Impact on health and mental health levels

The increase in nutritional levels of food for agencies would result from all three scenarios, increasing with the size of the enterprise and the number of agencies utilizing the products. Scenarios A and B (fixed kitchens) have room for growth and perhaps greater potential impact in this area. Access to a dietician or nutritionist, possibly through a partner organization in the partner space scenario, would increase the likelihood of positive nutritional outcomes in recipes.

Access to better food can have many positive health and mental health outcomes that are measureable (see Appendix A in *Finding Food* for some examples⁷). The FF Kitchen project, particularly the mobile kitchen scenario, offers important opportunities for social engagement (working as a team, processing with clients and volunteers at agencies). Social engagement has been shown to be an important factor in improving well-being and mental health. The advantage of the scenarios with larger staff (especially the Agency Coordinator position) is that more of these opportunities are likely to be taken if there is a staffperson designated to be responsive to agency needs.

⁷ Available at <http://parkdalecommunityeconomies.wordpress.com/2014/02/11/just-released-community-food-flow-final-research-report/>.

Long and short-term business viability

All new enterprises must navigate a balancing act between starting at a scale appropriate to their resources, and starting large enough to achieve goals and expand comfortably. The mobile kitchen alone sets limits to the amount that can be processed and the number of people who can participate at any time. In combination with a fixed kitchen, there is room for expansion. There is considerably less risk in the partner space scenario. Infrastructure and equipment upfront costs are reduced. The knowledge and expertise of the host organization can occasionally be accessed for support. The organizations can partner on events and joint fund-raising. Clients and volunteers can move between the two organizations. Sales can begin small and expenses can increase gradually with the sales.

The risk in a shared space comes later, when one organization becomes bigger and needs more space, or the host organization moves, merges or collapses. However, the higher risk in the long-term allows the new enterprise to achieve a stable footing, and one hopes, be able to weather some business vicissitudes.

Measurements of business viability include net surplus, revenue and sales figures, percentage of payroll to sales (this should be a clear target specific to the organization since the training aspect will make this a higher percentage than a standard business would use). Inventory turns (how long products are stored before selling) and other metrics can be applied as well. The measurement of long-term viability is more complex, and should include things like governance measurements:

- Is there clear decision-making process?
- Is the board well versed in their roles and responsibilities?
- Are they active in these roles?
- Is there a strategic plan that accounts for various levels of expansion, contraction and crisis?
- Is there a risk management system in place?
- Are the measurements for success clear to board and staff?
- Is there a stable staffing plan, with an HR committee, staff orientation/ retention procedures, and succession planning?

Funding options

The Food Flow project has identified a number of funding options for the start up. The team has been successful in a grant application for feasibility and business planning from the Toronto Enterprise Fund. The following table identifies additional possibilities for various aspects of the project. The feasibility study will allow partners to form agreements for moving forward and identify the best scenario for development. Secured funding can be matched with low interest community loans or bonds, or social finance loans to complete start up cost requirements.

Table 13: Funding opportunities

Project and project tasks	Funding (confirmed funding in blue; submitted in yellow)	Funding targets (estimated)	Potential partners or leads
Food Flow Kitchen			
Prefeasibility study, pilot development	Ontario Trillium Foundation (OTF)	2500	PARC, Daily Bread, West End Food Co-op
	Epicure Foundation	5000	PARC, Sistering, pilot community freezer
Business plan	TEF	10,000	PARC
Fund-raising	Ontario Trillium Foundation (OTF)	3500	PARC, Daily Bread, WEFC
Governance and partner development	McConnell (value chain, food hub funding) OTF (capacity building)? ONFC	10,000	Daily Bread, PARC, new social enterprise
Implementation	OTF	75,000	Daily Bread, new social enterprise
	TCF	20,000	WEFC
	Carrot Cache	5000	New social enterprise
	Growing Forward 2	65,000	Daily Bread, OFT, WEFC
	Metcalf		Daily Bread, PARC, new social enterprise, WEFC
	Alterna Learning Community grants (due June 2014)	5000	WEFC
	Anglican Community Development Fund	loans	St. Marks
	Atkinson	25,000 (no unsolicited)	PARC, Daily Bread, PCHC
	McLean (social enterprise) Metcalf	5000 30,000	PARC, Daily bread/ other partners/ West End Food Co-op
Evaluation, expansion	TCF Vital Ideas	30,000	PARC/ partners

Challenges and solutions

The Food Flow Development plan includes an assessment of risks and recommended steps to mitigate these risks. The following tables review this assessment for the Food Flow kitchen project.

Community-based processing at a commercial scale (producing product that can be purchased by or donated to the local agencies), was identified as an important solution in the Food Flow development process. It is also a scalable solution (if the space is sufficient); the equipment is relatively inexpensive and can be expanded gradually. Economies of scale may not apply until processing reaches a very large and mechanized scale. Community-scale processing can be used to process surplus donations as well as surplus product from farmers that they are not able to market (for cosmetic reasons, an over-supply, or other reasons).

The level of interest is very high (suggesting a likely success) and, at least in Parkdale, there is considerable capacity with the West End Food Co-op's cannery (developed jointly with PARC). The Food Flow project has strong partnerships that have helped in the advisory group since 2012, with fund-raising, specific expertise, space for meetings and other inputs. A steering committee specific to the project will be launched as the project progresses. The Food Flow Advisory Group members are listed in the “thank you” section at the front.

Risk factors and mitigation

The following tables provide analysis of the challenges and risks as well as potential solutions. Table 14.1 shows the risk factors for a commercial processing kitchen with an assessment of the impact on the project if the problem occurs, and the likelihood of the problem occurring. Table 14.2 reviews solutions to the risk factors, and identifies additional resources needed to provide these solutions.

Table 14.1: Risk factors

Risk factor	Impact of factor	Likelihood
High start-up costs and time	Major	Almost certain
New supply chains require new expertise and practices	Moderate	Almost certain
Interest in local may be a trend that passes	Moderate	Unlikely
Lack of knowledge of local food markets	Major	Almost certain
Food safety issues can prevent operations	Major	Possible
Economies of scale are not well understood for local processing	Major	Likely

Table 14.2: Risk mitigation options

Risk	Solutions	Resources needed
High start-up costs and time (major, almost certain)	- Access funding - Develop multiple funding streams - Identify development model that allows project to move forward in stages with partial funding - Source equipment through supportive suppliers (partial or full donations) - Access expertise to identify best equipment for project	- Fundraising staff/ contractors - Planning/ steering committee - Commercial kitchen expertise
New supply chains require new expertise and practices (moderate, almost certain)	- Engage supply chain expertise - Identify transportation solutions/ partners - Develop item list slowly; focus on similar items (dairy, or fruits and vegetables)	- Purchasing expertise - Transportation/ logistics expertise - Short item list requires distribution partner - Producer contracts

Interest in local may be a trend that passes (moderate, unlikely)	▪ Review and summarize existing research on local food market ▪ Consult experts on market trends	▪ Market research ▪ Sector-specific market analysis
Lack of knowledge of local food markets (major, almost certain)	▪ Develop marketing expertise ▪ Develop marketing strategies	▪ Marketing staff/team ▪ Funding for promo/marketing materials ▪ Customer lists
Food safety issues can prevent operations (major, possible)	▪ Access food safety requirements from Toronto Public Health ▪ Access Ontario materials (guidebook on processing)	▪ Staffing to review data and identify relevant regulations for project (chef expertise)
Economies of scale are not well understood for local processing (major, likely)	▪ Partner with Sustain Ontario or similar organization to develop this ▪ Contact similar processing projects ▪ Build detailed business plan with expansion details ▪ Identify target volumes and scale with steering committee	▪ Steering committee ▪ Partnerships for research ▪ Staff funding to support (or do) research

Development timeline

The Food Flow project has also provided a development timeline for all pilot projects, including the FF Kitchen (Table 15).

Table 15: Development timeline

April- June 2014		July-September 2014 ①		October- December 2014 ②		January- March 2015 ③		Goals by April 2015
Feasibility and planning (TEF and OTF funding)	Complete feasibility study	Build business plan including: ▪ Revenue/ expenses projections and targets ▪ Training and work placement goals ▪ Market need and recommendations ▪ Supply recommendations ▪ Staged development ▪ Governance structure	Develop-ment	▪ Implement governance structure ▪ Acquire funding and infrastructure ▪ Establish steering committee ▪ Engage chef expertise	Imple-mentation	▪ Begin staged implementation ▪ Prepare policies and procedures ▪ Begin outreach to agencies ▪ Link to portal, distribution partners ▪ Establish administration plan	Have established plan and majority of funding to implement in 2015	
	Submit additional funding (McConnell)							
			Complete partner development ▪ Develop access to space for project ▪ Develop partnerships for distribution, training		Develop partner agreements			

Recommendations

Although Year 1 revenue projections do not for the most part meet revenue requirements for the project, it is recommended that the project proceed. The projections are based on limited market assessment, and conservative predictions of initial sales. In addition, most food-based businesses do not break even in the first year or two, as food product margins are very tight. The business plan should account for an initial shortfall with other income streams (grants, loans, donors) and set targets for a self-sustaining operation that are reasonable and achievable. Full market assessment will identify the solutions and opportunities to bring revenues up to targets.

The new space scenario has the highest operating costs, and faces considerable risks stemming from the need for self-sufficiency at the outset. The mobile kitchen has the least operating cost and the quickest progress into positive net income. It is also the highest risk, as it represents an innovative and largely untested model for Ontario and Canada. The partner space model seems to have the best approach to risk and risk mitigation, mid-level costs, and some additional achievements of non-financial goals that recommend this scenario highly. It is recommended that project partners pursue the partner space but consider integrating the mobile kitchen into the more conventional model. This will mitigate many of the risks, and link the mobile kitchen to some of the other outcomes that are not possible in the small trailer space.

The project should proceed with more complete market assessment, business planning and governance structure development. Fund-raising for the initial year or two should be conducted now.

Next steps

1. Convene a decision-making group (steering committee)
2. Conduct strategic planning around three scenarios
3. Develop additional information as needed
4. Identify key partners
5. Submit fund-raising applications
6. Develop long-term governance plan and structure
7. Develop business plan⁸
8. Incorporate project if necessary

⁸ See North Dakota Department of Agriculture feasibility study for mobile processing unit for a list of specific planning needs for a mobile kitchen unit (N. C. Doty and Associates, 2009).

Appendix A: Sources for supply

1. Mobile kitchens

In the U.S.:

(From list of suppliers pp. 13-14 N. C. Doty and Associates. 2009. *Mobile Food Processing Unit Feasibility Study*. For North Dakota Department of Agriculture. July 2009.)

Coastal Concessions & Mobile Kitchens, Inc.
14108 US Hwy. 98, Lucedale, MS 39452
Phone: (601) 947-6160

Custom Concessions
104 Laurens Industrial Blvd., Dublin, GA 31021
Phone: (800) 910-8533

In Canada:
Mobile Kitchens Canada
3790 Davidson Road, Inverary, Ontario K0H 1X0
1-866-691-6462

2. Kitchen Equipment

Canada Food Equipment
45 Vansco Road, Etobicoke, ON M8Z 5Z8
Phone (416) 253-5100 Fax: (416) 253-2251

Nella Cutlery Toronto
148 Norfinch Drive Toronto, Ontario M3N 1X8
Phone: 416-740-2424 Fax: 416-740-9363

Hussmann Canada
5 Cherry Blossom Road Building 1, Unit 3 Cambridge, ON - N3H 4R7

Appendix B: Resources

Thanks to partners and supporters who provided discussion, input and hard facts for the development of this study, including:

- ❖ Daily Bread
- ❖ West End Food Co-op
- ❖ Parkdale Activity Centre
- ❖ The Food Flow Advisory Group
- ❖ Toronto Food Strategy/ Toronto Public Health

Other resources:

Barham, James, Debra Tropp, Kathleen Enterline, Jeff Farbman, John Fisk, and Stacia Kiraly. *Regional Food Hub Resource Guide*. U.S. Dept. of Agriculture, Agricultural Marketing Service. Washington, DC. April 2012.

Carter-Whitney and Miller. 2010. *Nurturing Fruit and Vegetable Processing in Ontario*. Metcalf Foundation Food Solutions papers

Christianson, Russ and M. Morgan. 2014. *Sudbury Local Food Hub Feasibility Study*. March 2014. Available at <http://www.eatlocalsudbury.com/#!/food-hub/c14ug>.

City of Toronto. 2008. *The State of Toronto's Food: Discussion Paper for a Toronto Food Strategy*. Toronto Public Health. May 2008 Revision.

D.C. Central Kitchen. Website. <http://www.dccentralkitchen.org/>

Daily Bread. Website. <http://www.dailybread.ca/programs/kitchen/daily-breads-storefront/#sthash.nH50U7rB.dpuf>.

Development Economics LLC. 2013. *A Feasibility Report for a Food Hub serving a 14 County Region of Northwest Michigan*. Presented to The McBain LDFA and Marilyn Knak. October 30, 2013. Available at foodsystems.msu.edu/uploads/.../Feasibility_Report_for_a_Food_Hub.pdf (PDF).

N. C. Doty and Associates. 2009. *Mobile Food Processing Unit Feasibility Study*. For North Dakota Department of Agriculture. July 2009.

Downtown Eastside Kitchen Table Project. Website. <http://dteskitchentables.org/>.

Fischer, M., Hamm, M., Pirog, R., Fisk, J., Farbman, J., & Kiraly, S. 2013. *Findings of the 2013 National Food Hub Survey*. Michigan State University Center for Regional Food Systems & The Wallace Center at Winrock International. September 2013. Retrieved from <http://foodsystems.msu.edu/activities/food-hub-survey>.

Foodlink (New York). Website. <http://foodlinkny.org>.

Kamizaki, Kuni. 2014. *Reassembling Community Food Flow: Community Food Distribution and Procurement in Toronto's West Central Neighbourhoods*. Available at <https://parkdalecommunityeconomies.wordpress.com/resources/the-community-food-project/>.

Miller, Sally. 2013. *Community Food Procurement in Toronto: Good Practices and Ideas for 2013*. Toronto Public Health/ Toronto Food Strategy. Unpublished report.

Miller, Sally. 2014. *Finding Food: Community Food Procurement in the City of Toronto*. Available at <https://parkdalecommunityeconomies.wordpress.com/resources/the-community-food-project/>.

National Good Food Network Food Hub Center. Website. <http://www.ngfn.org/resources/food-hubs>.

New City Market. N.D. *Draft Business Plan*. Available at <http://www.newcitymarket.org/draft-business-plan/>.

North Dakota Department of Agriculture. Website. Mobile Food Processing Unit. <http://www.nd.gov/ndda/program/mobile-food-processing-unit>.

Tse, Carmen and Valerie Tarasuk. 2008. "Nutritional assessment of charitable meal programmes serving homeless people in Toronto". *Public Health Nutrition*. 11(12), 1296-1305.

Appendix C: Price comparison of most common items

